

Task 1: Extended Grafana Dashboard

EECE 5155 – Wireless Sensor Networks and IoT Systems

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Overview

The seminar 4 panels dashboard using the `iot-sensors` InfluxDB bucket populated during the seminar via the MQTT publisher and bridge pipeline. All queries use Flux and target the IoT temperature dataset (Kaggle, 97,606 rows, Oct–Dec 2018).

Final Dashboard Screenshot

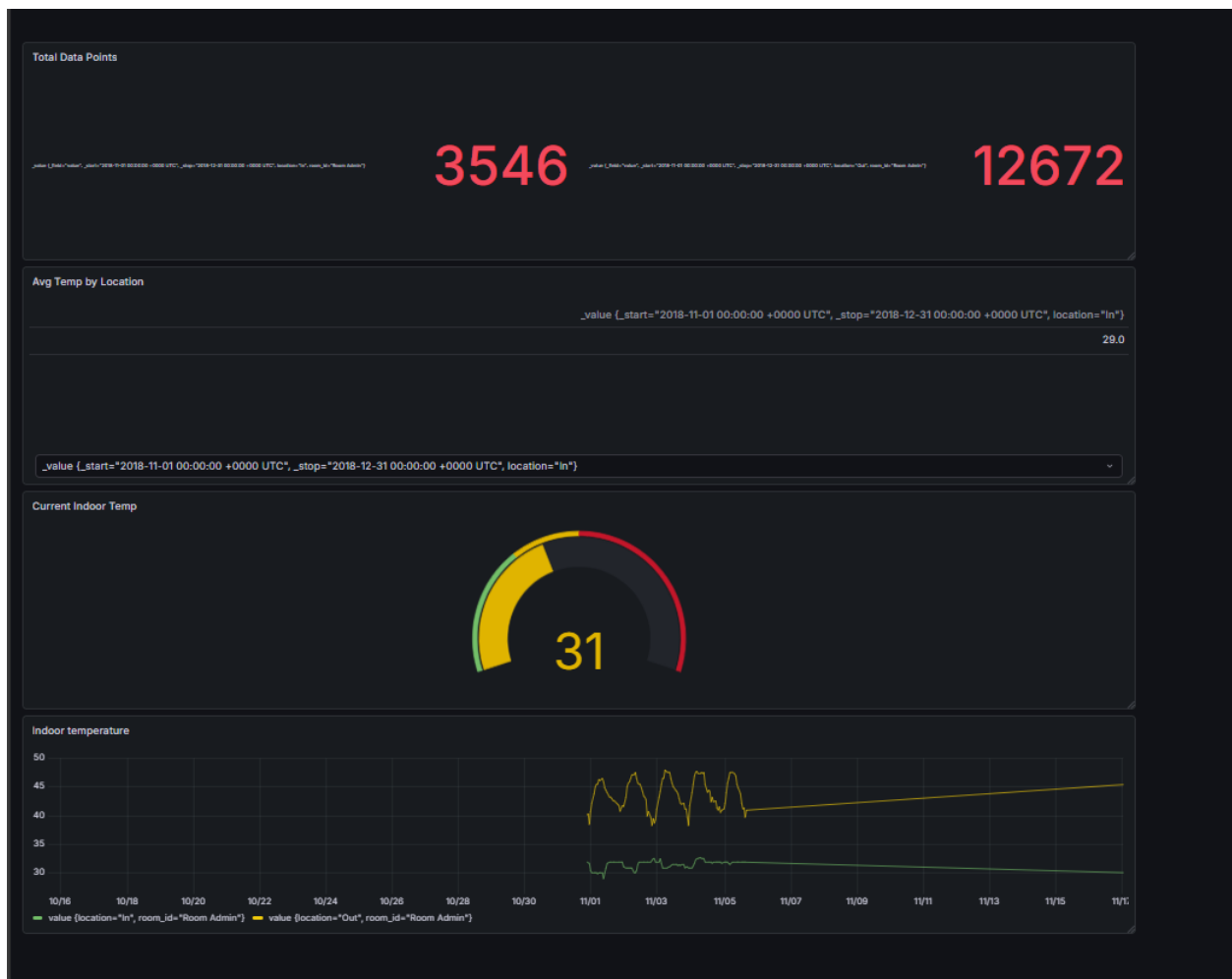


Figure 1: 4-panel Grafana dashboard – IoT Seminar, `iot-sensors` bucket

Panel Descriptions and Flux Queries

Panel 1 – Total Data Points (Stat)

Displays the total count of ingested readings split by indoor and outdoor location. Two Stat values are shown side by side: 3,546 (Indoor) and 12,672 (Outdoor).

Listing 1: Panel 1 – Indoor count

```
from(bucket: "iot-sensors")
  |> range(start: 2018-10-01T00:00:00Z, stop: 2018-12-31T00:00:00Z)
  |> filter(fn: (r) => r._measurement == "temperature")
  |> filter(fn: (r) => r._field == "value")
  |> filter(fn: (r) => r.location == "In")
  |> count()
```

Listing 2: Panel 1 – Outdoor count (second query)

```
from(bucket: "iot-sensors")
  |> range(start: 2018-10-01T00:00:00Z, stop: 2018-12-31T00:00:00Z)
  |> filter(fn: (r) => r._measurement == "temperature")
  |> filter(fn: (r) => r._field == "value")
  |> filter(fn: (r) => r.location == "Out")
  |> count()
```

Panel 2 – Avg Temp by Location (Table)

Displays the mean temperature for Indoor (29.0 °C) and Outdoor locations over the Nov–Dec 2018 window.

Listing 3: Panel 2 – Average temperature grouped by location

```
from(bucket: "iot-sensors")
  |> range(start: 2018-11-01T00:00:00Z, stop: 2018-12-31T00:00:00Z)
  |> filter(fn: (r) => r._measurement == "temperature")
  |> filter(fn: (r) => r._field == "value")
  |> group(columns: ["location"])
  |> mean()
```

Panel 3 – Current Indoor Temp (Gauge)

Shows the most recent indoor temperature reading as a colour-coded gauge. Thresholds: green <28 °C, yellow 28–35 °C, red ≥35 °C. Current reading: **31 °C** (yellow – warm but within tolerance). Threshold mode set to **Absolute** (not Percentage) to ensure correct colour mapping.

Listing 4: Panel 3 – Last indoor temperature reading

```
from(bucket: "iot-sensors")
  |> range(start: 2018-10-01T00:00:00Z, stop: 2018-12-31T00:00:00Z)
  |> filter(fn: (r) => r._measurement == "temperature")
  |> filter(fn: (r) => r._field == "value")
  |> filter(fn: (r) => r.location == "In")
  |> last()
```

Panel 4 – Indoor Temperature Over Time (Time Series)

Time-series chart showing hourly mean temperature for both Indoor (green, $\approx 30\text{--}32^\circ\text{C}$) and Outdoor (yellow, $38\text{--}47^\circ\text{C}$ with visible daily cycling Oct–Nov, stabilising after Nov 5). The outdoor oscillation pattern reflects natural diurnal variation; the flattening after Nov is an artefact of the aggregation window smoothing fewer data points.

Listing 5: Panel 4 – Hourly aggregated time series

```
from(bucket: "iot-sensors")
  |> range(start: 2018-10-15T00:00:00Z, stop: 2018-11-17T00:00:00Z)
  |> filter(fn: (r) => r._measurement == "temperature")
  |> filter(fn: (r) => r._field == "value")
  |> aggregateWindow(every: 1h, fn: mean)
```

AI Audit

What AI got right: Suggested a Stat panel using `count()` and a Table panel using `group(columns: ["location"]) |> mean()` – both were relevant and ran correctly after pasting into the Flux editor.

What AI got wrong / what I fixed: The initial queries used `range(start: -30d)` which returned no data because the dataset is from 2018. I changed the time range to a hardcoded absolute range (2018-10-01 to 2018-12-31). The gauge also displayed all red initially because Threshold mode defaulted to **Percentage** instead of **Absolute** – fixed in the panel options.

What I Learned

Grafana's `range(start: -30d)` is relative to *now*, so historical datasets return empty unless you hardcode absolute timestamps or set the dashboard time picker to match the data's actual date range. I also learned that gauge thresholds default to Percentage mode – at 28 and 35 on a 15–55 scale, percentage mode placed the boundaries at $\approx 26^\circ\text{C}$ and $\approx 29^\circ\text{C}$, collapsing the yellow band almost entirely and making 31°C appear red. Switching to Absolute fixed the colour mapping immediately. More broadly, visualisation type matters: the same mean-by-location query reads as a single number in Table view but becomes immediately comparable in Bar chart view – choosing the right panel type is as important as writing the correct query.